

4. IP Addressing Plan

VLSM applied within the assigned 10.49.0.0/16 block. Every subnet below is sized to the department's headcount after CR5's 25% growth is already added, so the existing addressing does not need to change if that growth occurs. Only 10.49.0.0/24 — a single /24 out of the full /16 — has been allocated; the remaining 255 /24-equivalent blocks (10.49.1.0 – 10.49.255.255) are held in reserve for growth beyond CR5, new departments, or additional sites, again without renumbering anything issued here.

4.1 Inside Addressing (10.49.0.0/24, VLSM)

Department	Current Users	Users +25% (CR5)	Network / Mask	Usable Hosts	Usable Range	Broadcast
Management	5	7	10.49.0.96/28	14	10.49.0.97 – 10.49.0.110	10.49.0.111
Finance	8	10	10.49.0.64/28	14	10.49.0.65 – 10.49.0.78	10.49.0.79
Creative & Design	20	25	10.49.0.0/27	30	10.49.0.1 – 10.49.0.30	10.49.0.31
Client Services & Sales	12	15	10.49.0.32/27	30	10.49.0.33 – 10.49.0.62	10.49.0.63
IT / Network Operations	5	7	10.49.0.80/28	14	10.49.0.81 – 10.49.0.94	10.49.0.95
Reception	3	4	10.49.0.112/29	6	10.49.0.113 – 10.49.0.118	10.49.0.119
Servers / DMZ	4 devices	4 devices	10.49.0.120/29	6	10.49.0.121 – 10.49.0.126	10.49.0.127
Mgmt/Native (switch SVIs)	—	reserve 14	10.49.0.128/28	14	10.49.0.129 – 10.49.0.142	10.49.0.143

4.2 Outside (Public/NAT) Addressing

Documentation-range addressing (RFC 5737 / TEST-NET-3 and TEST-NET-2) is used to simulate ISP-assigned public IPs for the Packet Tracer build, since real public IP space cannot be allocated in a lab environment.

Link	Network	ISP-side IP	Router-side IP
Primary WAN (R1 G0/0 – ISP1)	203.0.113.0/30	203.0.113.1 (ISP1)	203.0.113.2 (R1)
Backup WAN (R1 G0/1 – ISP2)	198.51.100.0/30	198.51.100.1 (ISP2)	198.51.100.2 (R1)